



Internship at Logpose Instruments

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Terahertz Optics and Lens Design

- Enabling New Materials & Applications (characterization of materials)
- Miniaturization for Portable Devices
- Signal Integrity for High-Speed Communications
- Precision for High-Resolution Imaging in Medical Application



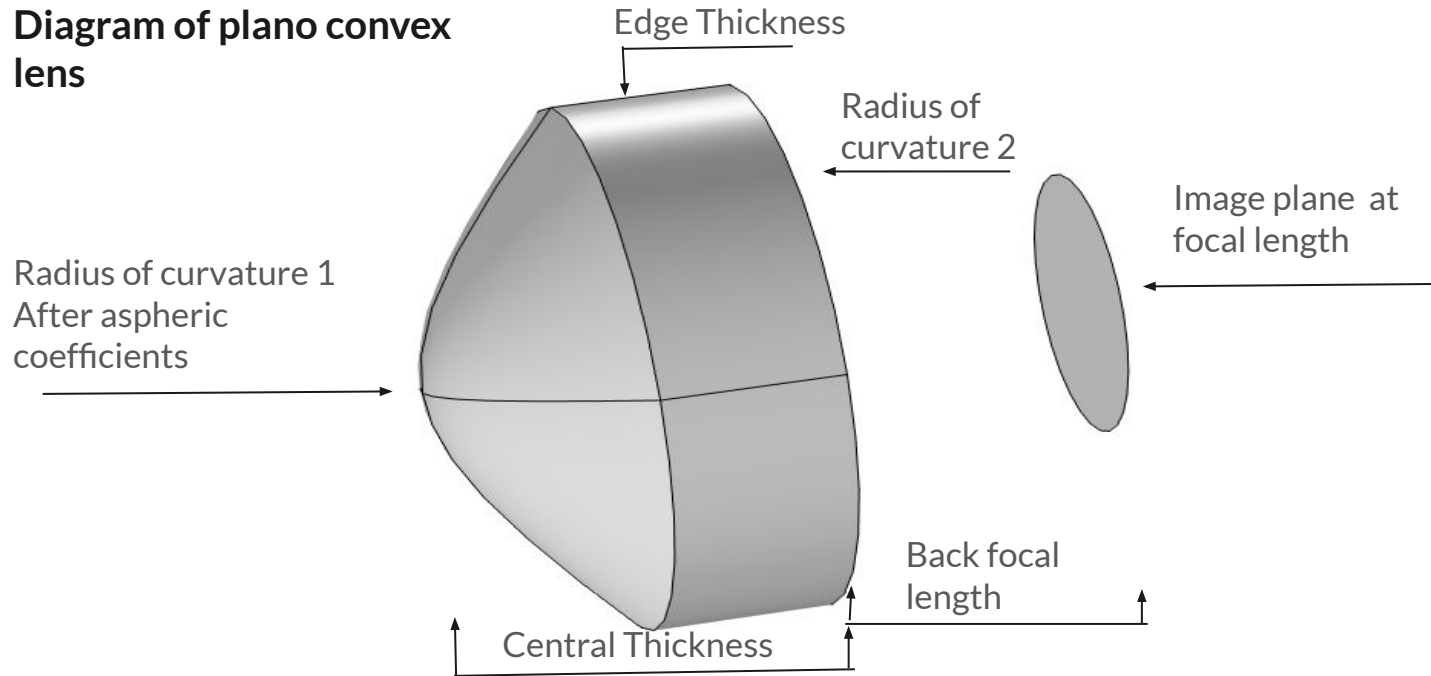
Lens Design: A brief overview

- Lens Design involves selection of various lens parameters like Radius of Curvatures of both the surfaces, aspheric coefficients, refractive index of lens, central thickness, wavelength of light etc
- Primarily, we adjust R and aspheric coefficients as rest of the parameters are known to us
- The parameters are chosen so as to minimize the spot radius and other defects

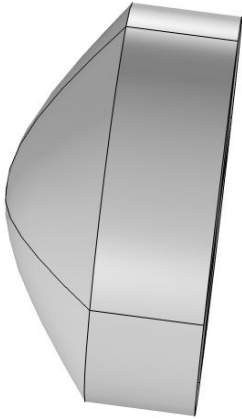
Described below is the aspheric equation

$$Z(y) = \frac{y^2/R}{1 + \sqrt{1 - (1 + K) \left(\frac{y}{R}\right)^2}} + A_2 y^2 + A_4 y^4 + A_6 y^6 + A_8 y^8 + A_{10} y^{10} + A_{12} y^{12} + \dots$$

Diagram of plano convex lens



Aspheric Coefficients: Adjust the sag of the lens (BFL= 10mm)



Putting second order
coefficient = $3\text{E-}2$

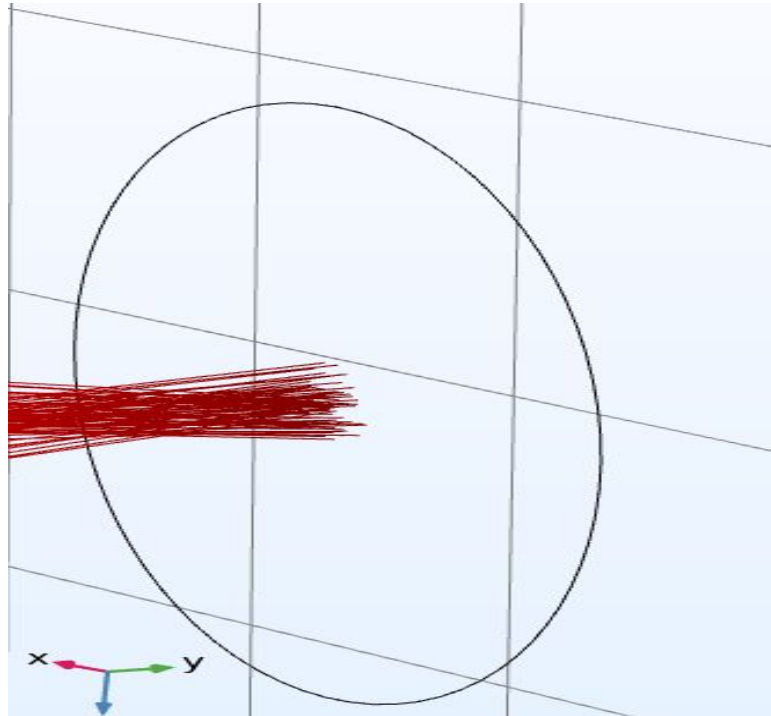
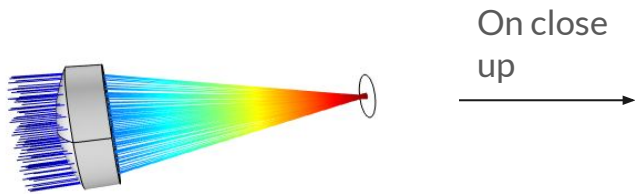




Spot Radius

- Throwing a collimated beam onto a convex surface ideally should result in a point focus
- But due to poor design/ unoptimized design beam fails to focus sharply
- This results in a circular spot instead of a point
- Spot Radius is largely affected by Radius of Curvature and aspheric coefficients
- Aspheric coefficients affect the curvature of lens to make it more uniform for all rays of light

Example: Spot





The earlier manual design method

- The previous design heavily involved manual calculations and trial and error by simulating on the COMSOL software
- In this method, the designer would manually enter radius of curvature of the lens and aspheric coefficients of the lenses one by one, and see which set of parameters gave the minimum spot radius
- This was laborious on the designer side, as well as there was no guarantee that the solution was optimal
- **There was no fixed method of tuning the parameters**



Towards a new automated design method

- To solve this problem, there was a need of automation of parameter tuning
- Vanilla COMSOL does not have any automation tools/ scripting methods
- As addons, COMSOL has interfacing capabilities with various softwares like AUTOCAD, Simulink, MATLAB etc
- MATLAB was the perfect choice for automating the simulations
- Next questions: what exactly to automate and is there an optimum way of finding the parameters



Initial Ideas

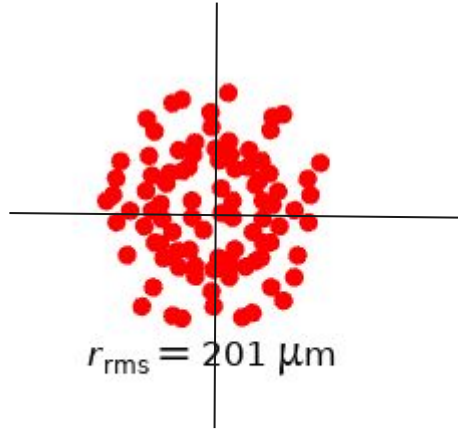
- Read 6-7 research papers
- Common themes: Use ML models paired with in built optimization tools for minimizing the defects
- ML models presented in papers: Genetic algorithms (GA), Inverse Neural Networks (INNs) and Nelder Mead Optimization (NMO)
- Brief overview of each of the methods:
 - **GA:** A population-based optimization technique inspired by natural selection and evolution
 - **INN:** Neural networks designed to infer input parameters from desired outputs
 - **NMO:** A derivative-free simplex method for minimizing nonlinear functions
- The biggest challenge in the current scenario: No data, and complex objective functions
- Almost all the methods used relied on some database or creating some simulations to generate initial data
- In this scenario : No previous dataset, as well as no idea how to generate an objective function for spot radius



Experimentation

- First needed to check the documentation for the MATLAB LiveLink interface in COMSOL
- Documentation was difficult to comprehend and did not mention anywhere how to find out the spot radius
- So first navigated the model methods and parameters to find how MATLAB interfaces with COMSOL
- After a lot of trial and error, found a method which returns the coordinates of the spot radius

The RMS spot radius



The RMS spot radius is calculated as follows wrt origin

$$R_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i^2 + y_i^2)}$$

MPHNAVIGATOR

mphnavigator - Model Object Navigator - COMSOL Multiphysics - Asphericlens_ray_optics_june30 (model)

File Tools Settings Help

Model Tree - SD_21.2

- COMSOL Model
 - batch
 - common
 - component
 - coordSystem
 - cpl
 - externalInterface
 - frame
 - func
 - geom
 - group
 - init
 - material
 - mesh
 - multiphysics
 - nodeGroup
 - param
 - spot1
 - physics
 - probe
 - result
 - pg1
 - rtrj1
 - vol1
 - pg2
 - spot1
 - param
 - dataset
 - dset1

Properties

	Property	Value	Allowed Value
1	additionallogicalex...	1	[String]
2	additionallogicalex...	off	on, off
3	arrangement	release	release, wavelength, single
4	circleactive	off	on, off
5	circleradiusexpr	1[um]	[String]
6	color	red	custom, black, blue, cyan, ...
7	columns	3	[Int]
8	customcolor	[1;0;0]	[DoubleArray]
9	data	ray1	none, parent, ray1, ip1
10	distribution	equalnumber	equalnumber, equalwidth
11	filterrelease	1	[Int]
12	filterreleaseactive	off	on, off

Copy set Copy get Copy table Copy

Methods

	Name	Value	Filter
1	active(boolean)		
2	author(String)		
3	author()		
4	axis()		
5	camera()		
6	clip()		
7	clip(String)		
8	comments(String)		

Copy call Copy

Path: model.result('pg2').feature('spot1') Copy

Coordinates of the spot

```
>> model.result('pg2').feature('spot1').getVertices(0,0)

ans =

2×331 single matrix

Columns 1 through 14

    0.0027    0.0018   -0.0002   -0.0002    0.0018    0.0001    0.0001    0.0011    0.0006    0.0008    0.0005    0.0008    0.0006    0.0011
    0.0005   -0.0008   -0.0011    0.0011    0.0007    0.0011   -0.0011   -0.0005   -0.0009   -0.0006    0.0011    0.0006    0.0009    0.0005

Columns 15 through 28

   -0.0007   -0.0009   -0.0006   -0.0009   -0.0007   -0.0010    0.0025    0.0044    0.0064    0.0054    0.0057    0.0056    0.0032    0.0049
    0.0009    0.0006    0.0011   -0.0006   -0.0009   -0.0072   -0.0067   -0.0054   -0.0031   -0.0033    0.0020    0.0028    0.0052    0.0022

Columns 29 through 42

   -0.0018   -0.0046   -0.0031   -0.0059   -0.0058   -0.0058   -0.0057   -0.0033   -0.0050    0.0015    0.0019    0.0069    0.0082    0.0093
    0.0072    0.0023    0.0051    0.0029    0.0021   -0.0020   -0.0028   -0.0052   -0.0022   -0.0124   -0.0099   -0.0097   -0.0107   -0.0060

Columns 43 through 56

    0.0105    0.0123    0.0108    0.0097    0.0097    0.0058    0.0013    0.0014   -0.0012   -0.0052   -0.0099   -0.0092   -0.0108   -0.0124
   -0.0022   -0.0015    0.0009    0.0055    0.0099    0.0094    0.0111    0.0124    0.0111    0.0103    0.0096    0.0054    0.0012    0.0014

Columns 57 through 70

   -0.0110   -0.0099   -0.0098   -0.0058   -0.0013    0.0018    0.0077    0.0096    0.0117    0.0170    0.0184    0.0247    0.0227    0.0231
   -0.0010   -0.0055   -0.0099   -0.0094   -0.0111   -0.0244   -0.0189   -0.0223   -0.0179   -0.0094   -0.0145   -0.0079   -0.0049    0.0035

Columns 71 through 84

    0.0233    0.0221    0.0144    0.0140    0.0078    0.0078    0.0023   -0.0080   -0.0069   -0.0146   -0.0159   -0.0216   -0.0229   -0.0231
    0.0004    0.0110    0.0150    0.0100    0.0004    0.0100    0.0004    0.0100    0.0000    0.0100    0.0155    0.0110    0.0000    0.0005
```



Model In Matlab

This shows exactly how the model was created in COMSOL

```
function out = model
%
% sample.m
%
% Model exported on Jul 1 2025, 22:52 by COMSOL 5.6.0.280.

import com.comsol.model.*
import com.comsol.model.util.*

model = ModelUtil.create('Model');

model.modelPath('D:\COMSOL_files');

model.component.create('comp1', true);

model.component('comp1').geom.create('geom1', 3);

model.component('comp1').mesh.create('mesh1');

model.component('comp1').physics.create('gop', 'GeometricalOptics', 'geom1');

model.study.create('std1');
model.study('std1').create('rtrac', 'RayTracing');
model.study('std1').feature('rtrac').activate('gop', true);

model.geom.load({'part1'}, 'Ray_Optics_Module\3D\Aspheric_Lenses\aspheric_even_lens_3d.mph', {'part1'});
model.component('comp1').geom('geom1').create('pi1', 'PartInstance');
model.component('comp1').geom('geom1').feature('pi1').set('selkeepnoncontr', false);
model.component('comp1').geom('geom1').feature('pi1').set('part', 'part1');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'R2', '0[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd0', '25.4.0[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd0', '25.4[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd1', '20[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd2', '20[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd1', '25.4[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd2', '25.4[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd1_clear', '20[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'd2_clear', '20[mm]');
model.component('comp1').geom('geom1').feature('pi1').setEntry('inputexpr', 'Tc', '11.2[mm]');
model.component('comp1').geom('geom1').feature.remove('pi1');
```

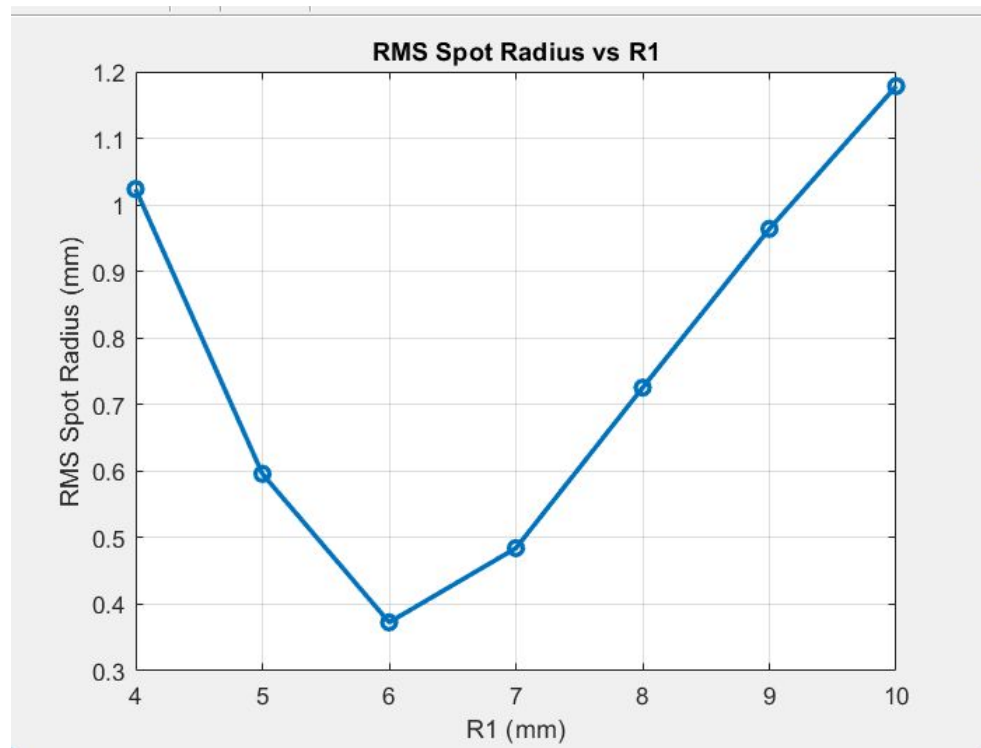


Devising the Method

- Saw the initial designs for the lenses first
- Noticed that there was major contribution to the minimum spot radius for one particular pair of Radius of Curvature and Aspheric Coefficient
- So first did a graphical analysis of how the spot radius varies with the Radius of Curvature
- And then a graphical analysis of how the spot radius varied with the aspheric coefficients using exponential scaling approach

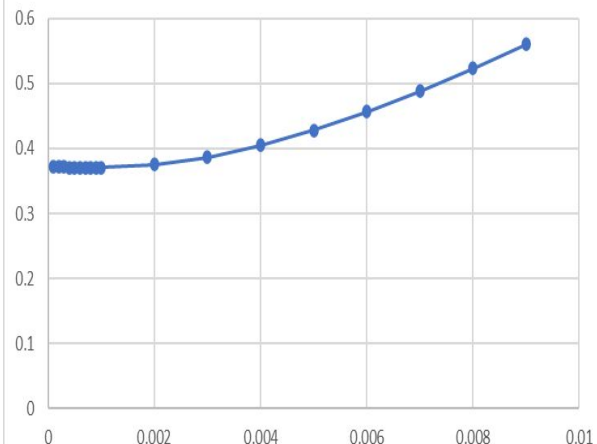
Graphs for BFL = 10 mm

All aspheric coefficients are zero

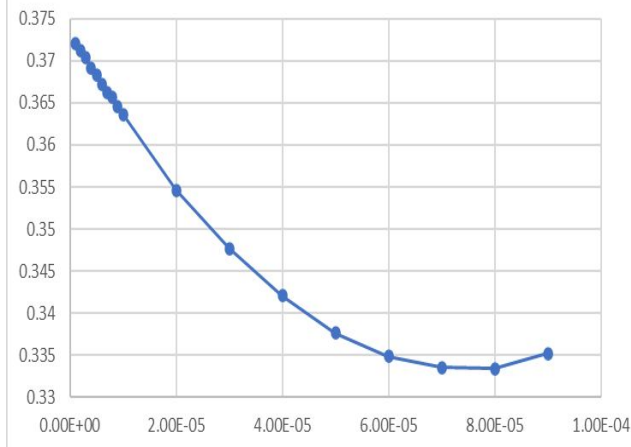


For $R_1 = 6$ mm

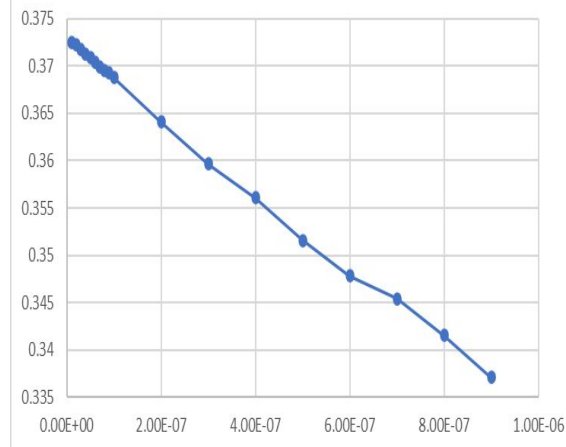
Spot radius vs A02_1



Spot Radius vs A04_1

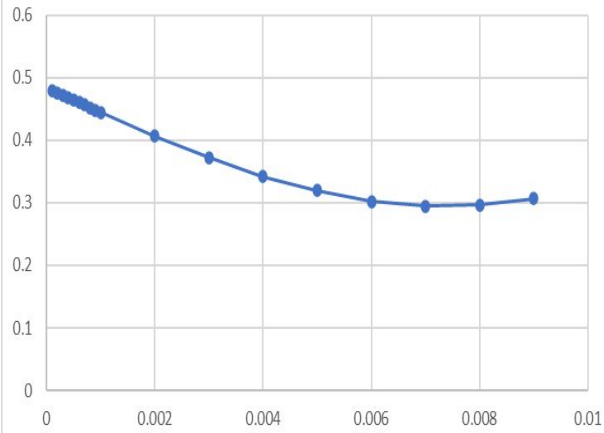


Spot Radius vs A06_1

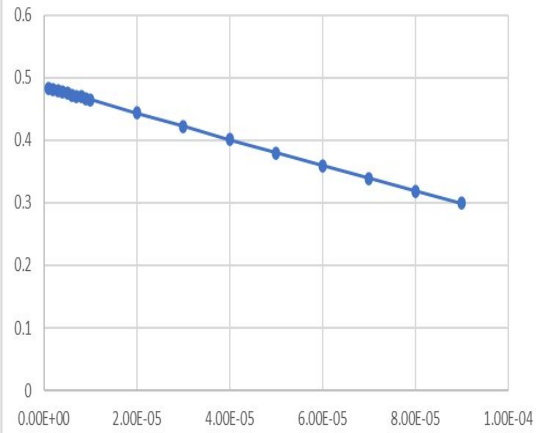


For $R_1 = 7$ mm

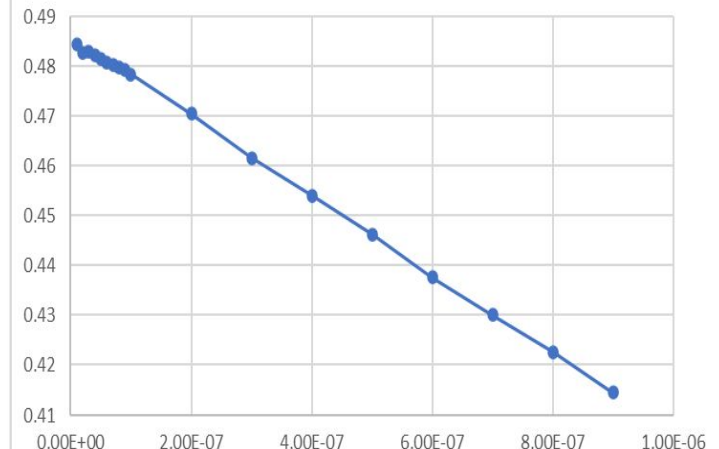
Spot Radius vs A02_1



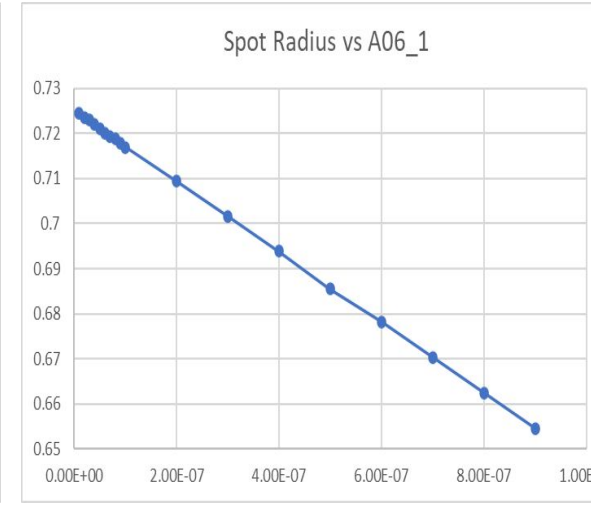
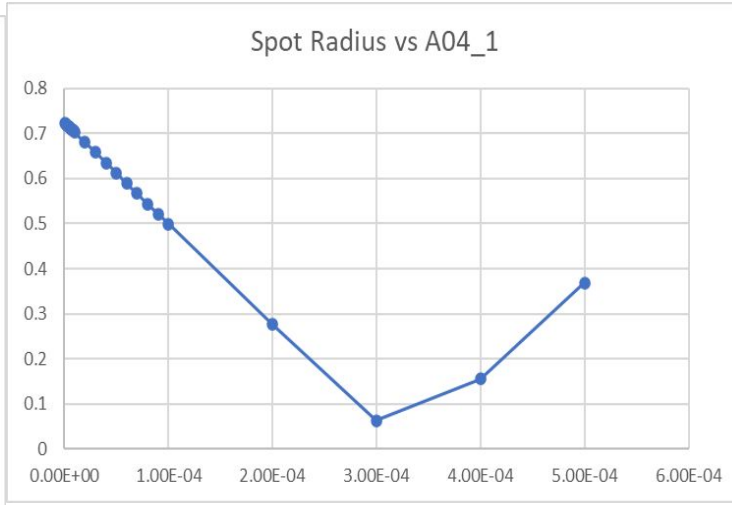
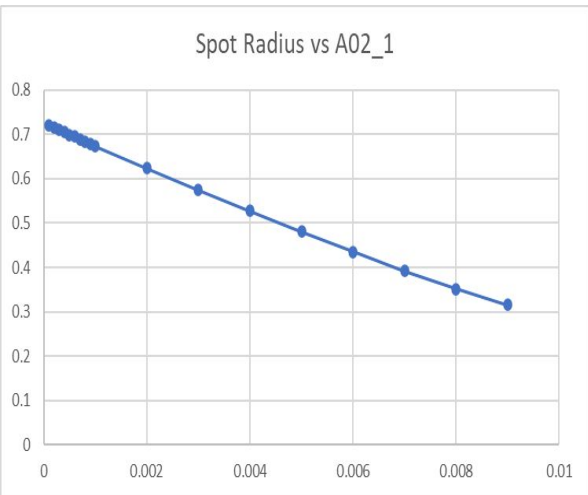
Spot Radius vs A04_1



A06_1 vs Spot Radius



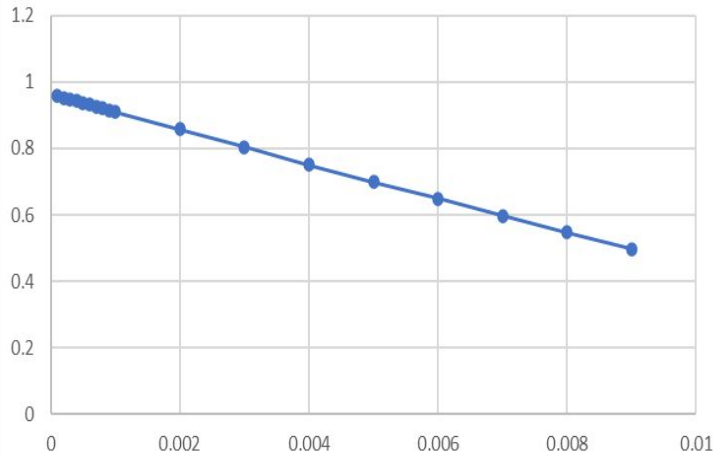
For R1= 8 mm



$\overline{A04_1} = 3.0e-04 \rightarrow \text{RMS} = 0.062556 \text{ mm}$

For $R_1 = 9 \text{ mm}$

Spot Radius vs A02_1





The Method

- First a graphical plot is generated which shows the variation of spot radius with radius of curvature of lens
- The points around and including the minimum are chosen
- The aspheric coefficients are exponentially varied for the chosen radius, first the growth of parameters is slow, then it increases by a factor of 10 and then by 100
- The range of aspheric coefficients is chosen so as they contribute uniquely to the optimization of the spot radius
- If edge thickness becomes negative, handle the exception and move to the next coefficient/Radius
- The nth coefficient's exponent varies from $[10^{(-n-2)}]$ to $[10^{(-n)}]$
- **Fast tuning:** A variation of this method in which the increase in new value from previous value, if it exceeds a certain threshold, then the process is halted and the next aspheric coefficient/ Radius of curvature is checked

$$T_e = T_c - z \left(\frac{d_0}{2} \right)$$



Example MATLAB output screen

```
A14_1 = 8.0e-15 --> RMS = 0.117568 mm
A14_1 = 9.0e-15 --> RMS = 0.116189 mm
A14_1 = 1.0e-14 --> RMS = 0.116345 mm
A14_1 = 2.0e-14 --> RMS = 0.114198 mm
A14_1 = 3.0e-14 --> RMS = 0.113237 mm
A14_1 = 4.0e-14 --> RMS = 0.109050 mm
A14_1 = 5.0e-14 --> RMS = 0.108023 mm
A14_1 = 6.0e-14 --> RMS = 0.119061 mm
A14_1 = 7.0e-14 --> RMS = 0.112163 mm
A14_1 = 8.0e-14 --> RMS = 0.110817 mm
[ERROR] A14_1 = 9.0e-14: Java exception occurred:
```

Note: All original designs R and Aspheric Coefficients were provided once the automated once were computed



Example: Fast Tuning of parameters (Threshold 1.05)

```
>> sweep_fast

--- R1 = 15.00 mm ---
Sweeping A02_1...
  A02_1 = 1.0e-04 --> RMS = 0.726823 mm
  A02_1 = 2.0e-04 --> RMS = 0.711157 mm
  A02_1 = 3.0e-04 --> RMS = 0.695158 mm
  A02_1 = 4.0e-04 --> RMS = 0.680184 mm
  A02_1 = 5.0e-04 --> RMS = 0.666453 mm
  A02_1 = 6.0e-04 --> RMS = 0.652620 mm
  A02_1 = 7.0e-04 --> RMS = 0.639227 mm
  A02_1 = 8.0e-04 --> RMS = 0.625894 mm
  A02_1 = 9.0e-04 --> RMS = 0.614906 mm
  A02_1 = 1.0e-03 --> RMS = 0.603449 mm
  A02_1 = 2.0e-03 --> RMS = 0.534615 mm
  A02_1 = 3.0e-03 --> RMS = 0.565358 mm
  [STEPPED INCREASE DETECTED] 0.534615 -> 0.565358
--> Results saved to R1_15.00_A02_1_sweep.csv
```


CSV File generation

 R1_6.00_A04_1_sweep.csv
 R1_6.00_A06_1_sweep.csv
 R1_6.00_A08_1_sweep.csv
 R1_6.00_A10_1_sweep.csv
 R1_6.00_A12_1_sweep.csv
 R1_6.00_A14_1_sweep.csv
 R1_7.00_A02_1_sweep.csv
 R1_13.00_A02_1_sweep.csv
 R1_13.00_A04_1_sweep.csv
 R1_13.00_A06_1_sweep.csv
 R1_13.00_A08_1_sweep.csv
 R1_13.00_A10_1_sweep.csv
 R1_13.00_A12_1_sweep.csv
 R1_13.00_A14_1_sweep.csv
 R1_14.00_A02_1_sweep.csv
 R1_14.00_A04_1_sweep.csv
 R1_14.00_A06_1_sweep.csv
 R1_14.00_A08_1_sweep.csv
 R1_14.00_A10_1_sweep.csv
 R1_14.00_A12_1_sweep.csv

A04_1	RMS_Radius_mm			
1.00E-06	0.37207			
2.00E-06	0.37117			
3.00E-06	0.37036			
4.00E-06	0.36921			
5.00E-06	0.36828			
6.00E-06	0.36722			
7.00E-06	0.36628			
8.00E-06	0.36569			
9.00E-06	0.36452			
1.00E-05	0.36362			
2.00E-05	0.35462			
3.00E-05	0.34767			
4.00E-05	0.34203			
5.00E-05	0.33763			
6.00E-05	0.33482			
7.00E-05	0.33356			
8.00E-05	0.33337			
9.00E-05	0.33522			
0.0001	NaN			
0.0002	NaN			
0.0003	NaN			
0.0004	NaN			
0.0005	NaN			
0.0006	NaN			
0.0007	NaN			

 R1_6.00_A04_1_sweep 



Results: BFL = 10 mm

Earlier Design

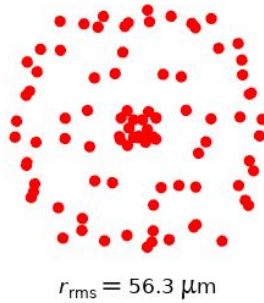
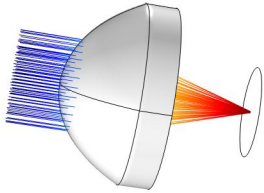
R	8.046
A02_1	1.5E-5
A04_1	3.098E-4
A06_1	3.15E-8
A08_1	1E-12

Optimized Design

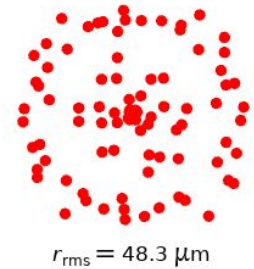
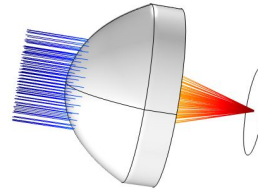
R	8.0015
A02_1	0
A04_1	3.069E-4
A06_1	0
A08_1	0

The spot radii

Manual Design



Optimized Design



Spot Radius Improvement
14.2 %

Time taken to for script termination: ~ 12 hours (Default mode)



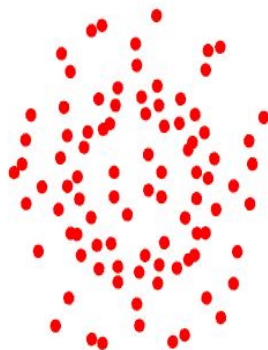
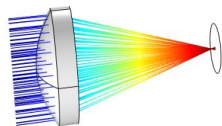
Results: BFL = 32.5 mm

R	17.381
A02_1	1E-3
A04_1	3.186E-5
A06_1	1E-9
A08_1	5E-12

R	17
A02_1	0
A04_1	3E-5
A06_1	0
A08_1	0

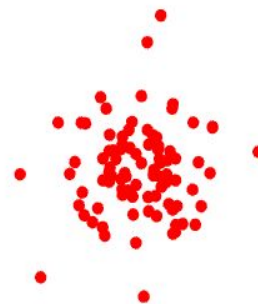
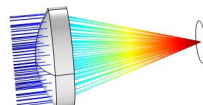
The spot radii

Manual Design



$$r_{\text{rms}} = 201 \mu\text{m}$$

Optimized Design



$$r_{\text{rms}} = 44.8 \mu\text{m}$$

Spot Radius Improvement

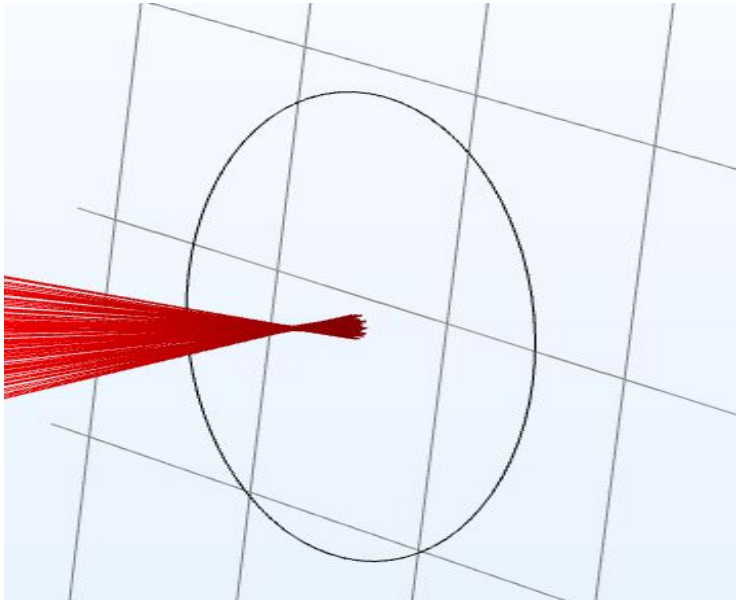
77.2 %

Time taken to for script termination: ~ 4 hours (Fast mode)

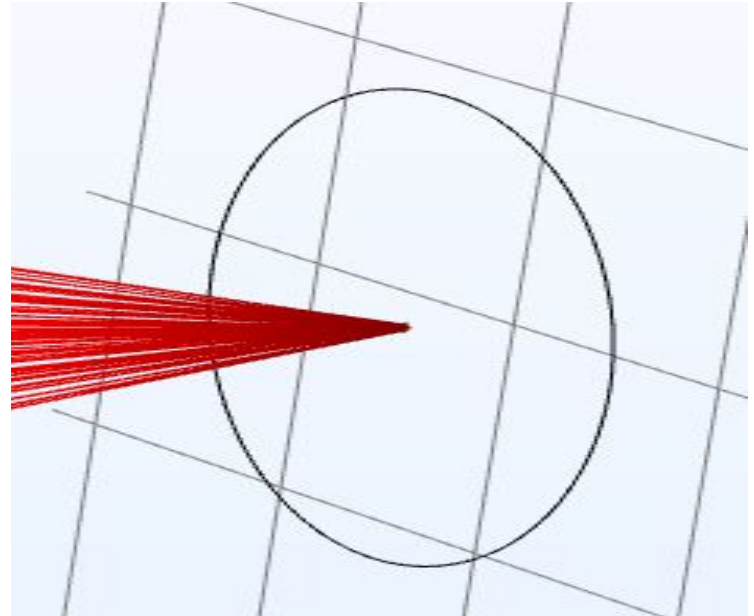


On zooming

Manual Design



Optimized Design



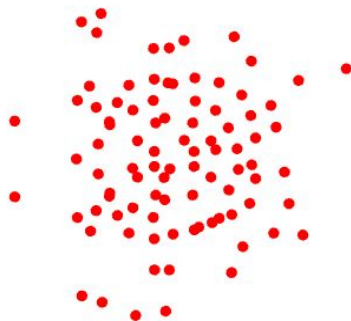
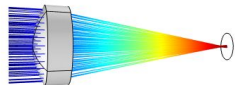


Results: BFL= 50mm

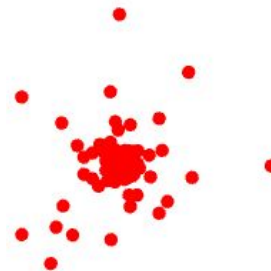
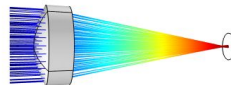
R	24.536
A02_1	1E-6
A04_1	1.2E-5
A06_1	1E-12
A08_1	5E-15

R	25
A02_1	0
A04_1	1E-5
A06_1	0
A08_1	0

The Spot Radii



$$r_{\text{rms}} = 286 \mu\text{m}$$



$$r_{\text{rms}} = 111 \mu\text{m}$$

Spot Radius Improvement

61.2 %

Time taken to for script termination: ~ 4 hours (Fast mode)



Observations

- The initial radius from the radius sweep which gives minimum spot radius does not always correspond to the final radius of curvature, it is possible that the radius occurs ahead of the minima as well
- The major contribution to the minimization of spot radius is from a single aspheric coefficient for a given Radius of Curvature
- The precision of the parameters does not significantly affect the spot radius
- Nelder Mead Optimization alone cannot give the optimum result, as it is susceptible to getting trapped in local minima
- It is infallible to optimize all parameters at once with constrained optimization, as the step sizes are not properly defined in fmincon



Cntd...

- The functions of spot radius with respect to aspheric coefficients keeping Radius of curvature fixed are either monotonically increasing or decreasing or are sharply concave upward in nature at point of minima
- Here x is the aspheric coefficient and $f(x)$ is the spot radius as a function of aspheric coefficient
- Optimum radius of curvatures occurred 2-3 mm ahead of the initial guess of radius of curvature

Concave upward: $f''(x) > 0$

- Monotonically increasing:

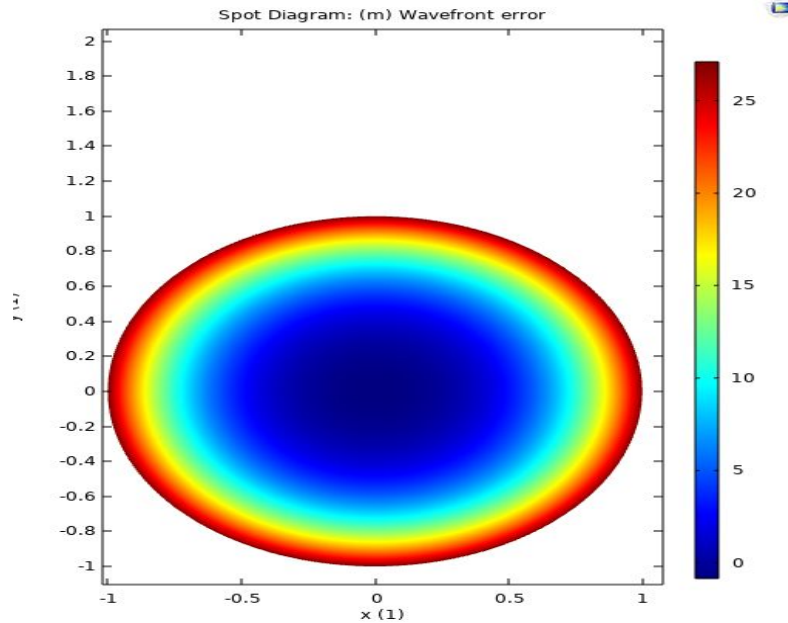
$$f'(x) \geq 0 \quad \forall x \in D$$

- Monotonically decreasing:

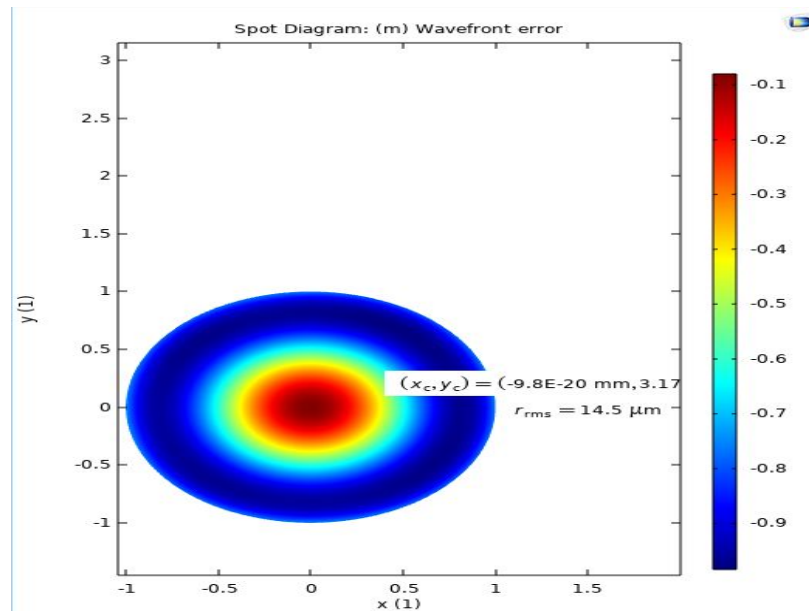
$$f'(x) \leq 0 \quad \forall x \in D$$

Spherical Aberrations

Unoptimized



Optimized





Summary of Contributions

- Interfacing COMSOL and MATLAB with LiveLink
- Figuring out how to automatically set and get model parameters
- Logic of placing of image plane at focal point, finding the spot coordinates and deriving rms spot radius from the coordinates
- Developing the automated parameter tuning method in MATLAB with the following USPs:
 - Exponential and systematic scaling of parameters in viable range
 - Fast mode of parameter tuning
 - Graphical analysis
 - Dataset Contribution
- Literature Reviews of existing approaches and exploring Nelder Mead Optimization
- Figuring out how to plot spherical aberrations and understanding zernike polynomials



Future Scope

- **Design of a better UX**
- Figuring out mathematical models to represent the various defects for application of gradient based algorithms
- Sourcing or generating datasets for preparing ML models
- Handling defects other than spot radius like spherical aberrations and wave optics defects
- More deterministic and faster algorithms for efficient prototyping



Conclusion

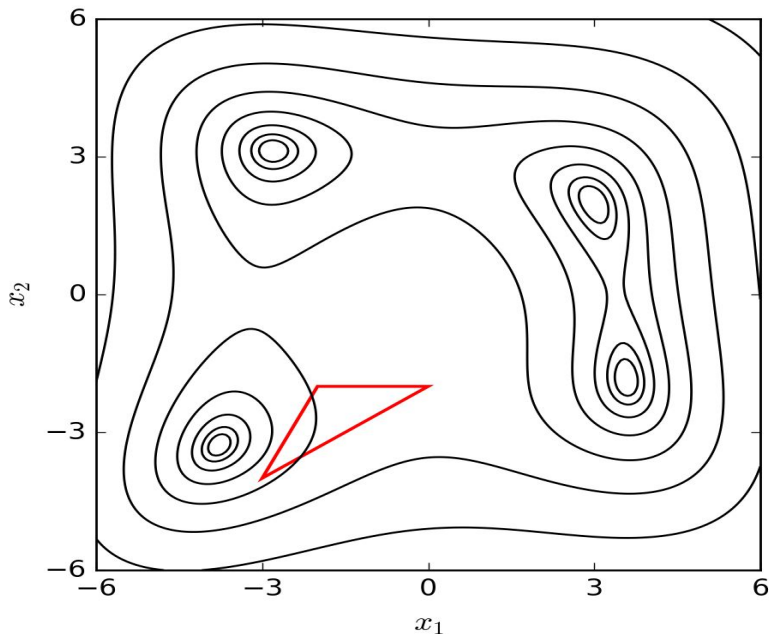
- Experimenting with computational physics hands-on in COMSOL
- Interfacing of softwares
- Creating parameter tuning models
- Understanding functional and non-functional requirements of softwares
- Data Analysis
- Testing out optimization methods
- Scientific presentation



Thank You

Open for questions!

Extras: The Nelder Mead Optimization Method



Derivative Free simplex based method

Derivative-Free Optimization: Nelder-Mead is a heuristic search algorithm that does not require gradient information, making it suitable for optimizing non-differentiable or noisy functions.

Simplex-Based Method: It uses a simplex (a polytope of $n+1$ vertices in n -dimensional space) which evolves over iterations through operations like reflection, expansion, contraction, and shrinkage.

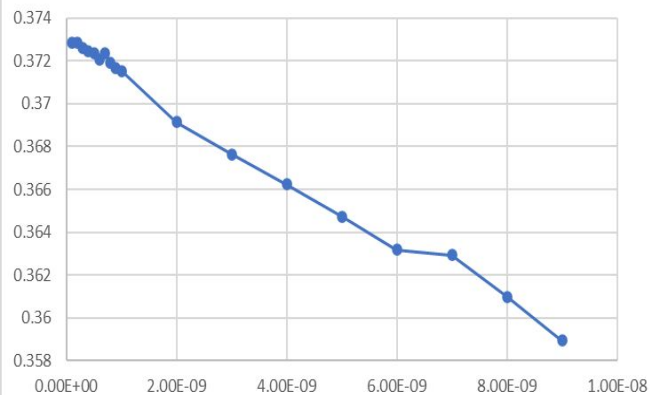
Local Optimization: The method is efficient for local optimization in low-dimensional spaces, but it may converge to local minima and is not guaranteed to find global optima.

Widely Used for Tuning: Due to its simplicity and robustness, it is frequently used in experimental tuning, engineering design problems, and cases where function evaluations are expensive.

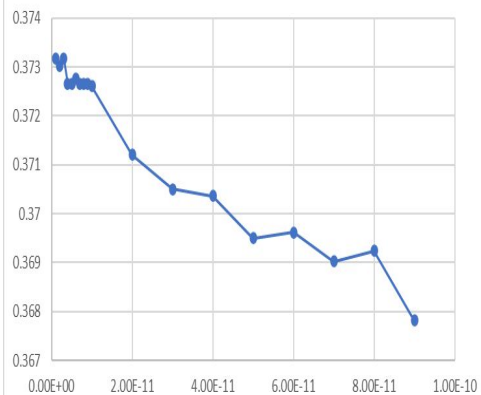
Susceptible to getting trapped in local minima

Extras: Higher order aspheric coefficients for R1 = 6mm

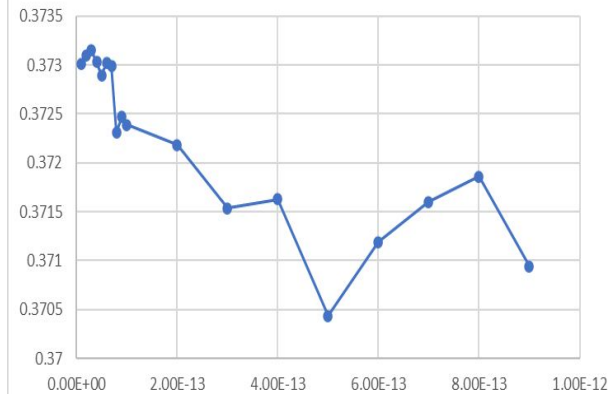
Spot Radius vs A08_1



Spot Radius vs A10_1



Spot Radius vs A12_1



Extras: Plots

